Dec PDP10/KS10 Simulator Usage

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1. Introduction

Originally the DEC PDP10 computer started as the PDP6. This was a 36 bit computer that was designed for timesharing, which was introduced in 1964. The original goal of the machine was to allow for processing of many 6 bit characters at a time. 36 bits were also common in machines like the IBM 7090, GE 645, and Univac 11xx lines. Several systems influenced the design of the PDP6, like CTSS, Lisp, support for larger memory. The PDP6 was canceled by DEC due to production problems. The engineers designed a smaller replacement, which happened to be a 36 bit computer that looked very much like the PDP6. This was called the PDP10. Later renamed to "DECSystem-10". The system supported up to 256K words of memory.

The first PDP10 was labeled KA10, and added a few instructions to the PDP6. Both the PDP6 and PDP10 used a base and limit relocation scheme. The KA10 generally offered two registers, one of user data and the second for user shared code. These were referred to the Low-Segment and High-Segment, the High-Segment could be shared with several jobs. The next version was called KI10 for Integrated. This added support for paging and double precision floating point instructions. It also added 4 sets of registers to improve context switching time. It could also support up to 4Mega words of memory. Following the KI10 was the KL10 (for Low-Cost). The KL10 added double precision integer instructions and instructions to improve COBOL performance. This was the first version which was microcoded. The KL10 was extended to support user programs larger then 256k. The final version to make it to market was the KS10 (for Small), this was a bit-slice version of the PDP10 which used UNIBUS devices which were cheaper then the KL10 devices.

The original operating system for the PDP6/PDP10 was just called "Monitor". It was designed to fit into 6K words. Around the 3rd release Swapping was added. The 6'th release saw the addition of virtual memory. Around the 4'th release it was given the name "TOPS-10". Around this time BBN was working on a paging system and implemented it on the PDP10. This was called "Tenex". This was later adopted by DEC and became "Tops-20".

During the mid 60's a group at MIT, who where not happy with how Multics was being developed decided to create their own operating system which they called Incompatible Timesharing System "ITS", was a play on the original project called Compatible Timesharing System "CTSS". CTSS was implemented by MIT on their IBM 7090 as an experimental system that allowed multiple time sharing users to co-exist on the same machine running batch processing. Hence the term Compatable.

Also during the mid 60's a group at Stanford Artificial Intelligence Laboratory (SAIL), started with a version of TOPS-10 and heavily modified it to become WAITS.

During the 70's Tymshare starting with DEC TOPS-10 system modified it to support random access files, paging with working sets and spawn-able processes. This ran on the KI10, KL10 and KS10.

The PDP10 was ultimately replaced by the VAX.

2. Simulator Files

To compile the DEC 10/KS10 simulator, you must define USE_INT64 as part of the compilation command line.

Subdirectory	*File*	*Contains*
PDP10	kx10_defs.h	KL10 simulator definitions
	kx10_cpu.c	KL10 CPU.
	kx10_disk.h	Disk formatter definitions
	kx10_disk.c	Disk formatter routine.
	kx10_sys.c	KS10 System interface
	kx10_rp.c	RH10/RH20 Disk controller, RP04/5/6 Disks.
	kx10_tu.c	RH10/RH20 TM03 Magnetic tape controller.
	kx10_rh.c	RH11 Controller.
	ks10_lp.c	LP 10 Line printer.
	ks10_dz.c	DZ11 communications controller.
	ks10_cty.c	KS10 front panel.
	ks10_uba.c	KS10 Unibus interface.
	kx10_imp.c	IMP11 interface to ethernet.
	ks10_ch11.c	Chaosnet 11 interface.
	ks10_dup.c	DUP11 interface.
	ks10_dup.h	Header file for DUP11 interface
	pdp11_ddcmp.h	Header to define interface for Decnet.
	ks10_kmc.c	KMC communications controller.

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3. KS10 Features

The KS10 simulator is configured as follows:

Device Name(s)	*Simulates*
CPU	KS10 CPU with 256KW of memory
CTY	Console TTY.
LP20	LP10 Line Printer
RPA	RH10/RH20 Disk controllers via RH10
TUA	TM02 Tape controller via RH10/RH20
DZ	DC10 Communications controller.
IMP	IMP network interface
CH	CH10 Chaosnet interface
DUP	DUP11 interface.
KDP	KMC11 interface.

3.1 CPU

The CPU options include setting memory size and O/S customization.

SET CPU 256K	Sets memory to 256K
SET CPU 512K	Sets memory to 512K
SET CPU 1024K	Set memory to 1024K
SET CPU ITS	Adds ITS Pager and instruction support to ITS KS
SET CPU NOIDLE	Disables Idle detection
SET CPU IDLE	Enables Idle detection

CPU registers include the visible state of the processor as well as the control registers for the interrupt system.

Name	Size	Comments	OS Type
PC	18	Program Counter	

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Name	Size	Comments	OS Type
FLAGS	18	Flags	
FM0-FM17	36	Fast Memory	
SW	36	Console SW Register	
MI	36	Monitor Display	
PIR	8	Priority Interrupt Request	
PIH	8	Priority Interrupt Hold	
PIE	8	Priority Interrupt Enable	
PIENB	1	Enable Priority System	
BYF5	1	Byte Flag	
UUO	1	UUO Cycle	
NXM	1	Non-existing memory access	
CLK	1	Clock interrupt	
OV	1	Overflow enable	
FOV	1	Floating overflow enable	
APRIRQ	3	APR Interrupt number	
UB	22	User Base Pointer	
EB	22	Executive Base Pointer	
FMSEL	8	Register set selection	
SERIAL	10	System serial number	
PAGE_ENABLE	1	Paging system enabled	
PAGE_FAULT	1	Page fault flag.	
FAULT_DATA	36	Page fault data	
SPT	18	Special Page table	
CST	18	Memory Status table	
PU	36	User data	
CSTM	36	Status Mask	

The CPU can maintain a history of the most recently executed instructions.

This is controlled by the SET CPU HISTORY and SHOW CPU HISTORY commands:

SET CPU HISTORY	clear history buffer
SET CPU HISTORY=0	disable history
SET CPU HISTORY=*n*	enable history, length = n
SHOW CPU HISTORY	print CPU history
SHOW CPU HISTORY=*n*	print first n entries of CPU history

Instruction tracing shows the Program counter, the contents of AC selected, the computed effective address. AR is generally the contents the memory location referenced by EA. RES is the result of the instruction. FLAGS shows the flags after the instruction is executed. IR shows the instruction being executed.

3.2 KS10 front end processor

3.2.1 Console Teletype (CTY)

The console station allows for communications with the operating system.

SET CTY 7B	7 bit characters, space parity.
SET CTY 8B	8 bit characters, space parity.
SET CTY 7P	7 bit characters, space parity.
SET CTY UC	Translate lower case to upper case.

The CTY also supports a method for stopping TOPS10 operating system.

3.3 Unit record I/O Devices

3.3.1 LP20 printer (LP20)

The line printer (LP20) writes data to a disk file as ASCII text with terminating newlines. Currently set to handle standard signals to control paper advance. Currently only one line printer device is supported.

SET LP20 LC	Allow printer to print lower case
SET LP20 UC	Print only upper case
SET LP20 LINESPERPAGE=*n*	Sets the number of lines before an auto form feed is generated. There is an automatic margin of 6 lines. There is a maximum of 100 lines per page.

SET LP20 NORMAL	Allows the VFU to be loading into the printer.
SET LP20 OPTICAL	VFU is fixed in the printer.
SET LP20 ADDR=value	Sets the Unibus address of device. (default: 775400)
SET LP20 VECT=value	Sets the interrupt vector of device. (default: 754)
SET LP20 BR=value	Sets the interrupt priority of device. (default: 5)
SET LP20 CTL=value	Sets the Unibus adapter number. (default: 3)

3.4

These characters control the skipping of various number of lines.

Character	Effect
014	Skip to top of form.
013	Skip mod 20 lines.
020	Skip mod 30 lines.
021	Skip to even line.
022	Skip to every 3 line
023	Same as line feed (12), but ignore margin.

3.5 Disk I/O Devices

The PDP10 supported many disk controllers. The KS10 supports only mass bus devices.

3.6 Massbus Devices

Massbus devices are attached via RH11's.

3.6.1 RP Disk drives

The KS10 supports one RH11 controller with up to 8 RP drives can be configured. These are addresses as RP Disks can be stored in one of several file formats, SIMH, DBD9 and DLD9. The later two are for compatibility with other simulators.

SET RPA*n* RP04	Sets this unit to be an RP04.(19MWords)
SET RPA*n* RP06	Sets this unit to be an RP06.(39MWords)
SET RPA*n* RP07	Sets this unit to be an RP07.(110MWords)
SET RPA*n* LOCKED	Sets this unit to be read only.
SET RPA*n* WRITEENABLE	Sets this unit to be writable.
SET RPA ADDR=value	Sets the Unibus address of device. (default: 776700)
SET RPA VECT=value	Sets the interrupt vector of device. (default: 254)
SET RPA BR=value	Sets the interrupt priority of device. (default: 6)
SET RPA CTL=value	Sets the Unibus adapter number. (default: 1)

To attach a disk use the ATTACH command:

ATTACH RP*An* *<file>*	Attaches a file
ATTACH RA*n* -f *<format>* *<file>*	Attaches a file with the given format.
ATTACH RPA*n* -n <file>	Creates a blank disk.

Format can be SIMH, DBD9, DLD9.

3.6.2 TUA Tape drives.

The TU is a Mass bus tape controller using a TM03 formatter.

SET TUA*n* LOCKED	Sets the mag tape to be read only.
SET TUA*n* WRITEENABLE	Sets the mag tape to be writable.
SET TUA ADDR=value	Sets the Unibus address of device. (default: 772440)
SET TUA VECT=value	Sets the interrupt vector of device. (default: 224)
SET TUA BR=value	Sets the interrupt priority of device. (default: 6)
SET TUA CTL=value	Sets the Unibus adapter number. (default: 3)

3.7 Terminal Multiplexer I/O Devices

All terminal multiplexers must be attached in order to work. The ATTACH command specifies the port to be used for Telnet sessions:

ATTACH <device> *<port>* set up listening port
where port is a decimal number between 1 and 65535 that is not being used other TCP/IP activities.

Once attached and the simulator is running, the multiplexer listens for connections on the specified port. It assumes that any incoming connection is a Telnet connections. The connections remain open until disconnected either by the Telnet client, a SET <device> DISCONNECT command,or a DETACH <device> command.

SET <device> DISCONNECT=*n* Disconnect line *n*
The <device> implements the following special SHOW commands

SHOW <device> CONNECTIONS	Displays current connections to the <device>
SHOW <device> STATISTICS	Displays statistics for active connections
SHOW <device> LOG	Display logging for all lines.

Logging can be controlled as follows:

SET <device> LOG=*n*=*filename*	Log output of line *n* to *filename*
SET <device> NOLOG	Disable logging and close log file

3.7.1 DZ11 Terminal Controller

The DZ device was the standard terminal multiplexer for the KS10. Lines came in groups of 8.

A max of 32 lines can be supported.

SET DZ LINES=*n*	Set the number of lines on the DZ10, multiple of 8.
SET DZ ADDR=value	Sets the Unibus address of device. (default: 760000)
SET DZ VECT=value	Sets the interrupt vector of device. (default: 340)
SET DZ BR=value	Sets the interrupt priority of device. (default: 5)
SET DZ CTL=value	Sets the Unibus adapter number. (default: 3)

3.8 Network Devices.

3.8.1 IMP Interface Message Processor

This allows the KS to connect to the Internet. Currently only supported under ITS. ITS and other OS that used the IMP did not support modern protocols and typically required a complete rebuild to change the IP address. Because of this the IMP processor includes built in NAT and DHCP support. For ITS the system generated IP packets which are translated to the local network. If the HOST is set to 0.0.0.0 there will be no translation. If HOST is set to an IP address then it will be mapped into the address set in IP. If DHCP is enabled the IMP will issue a DHCP request at startup and set IP to the address that is provided. DHCP is enabled by default.

SET IMP MAC=*xx:xx:xx:xx:xx:xx*	Set the MAC address of the IMP to the value given.
SET IMP IP=*ddd.ddd.ddd.ddd/dd*	Set the external IP address of the IMP along with the net mask in bits.
SET IMP GW=*ddd.ddd.ddd.ddd*	Set the Gateway address for the IMP.
SET IMP HOST=*ddd.ddd.ddd.ddd*	Sets the IP address of the PDP10 system.
SET IMP DHCP	Allows the IMP to acquire an IP address from the local network via DHCP. Only HOST must be set if this feature is used.
SET IMP NODHCP	Disables the IMP from making DHCP queries
SET IMP ARP=*ddd.ddd.ddd.ddd=xx:xx:xx:xx:xx:xx*	Creates a static ARP entry for the IP address with the indicated MAC address.
SET IMP ADDR=value	Sets the Unibus address of device. (default: 767600)
SET IMP VECT=value	Sets the interrupt vector of device. (default: 250)
SET IMP BR=value	Sets the interrupt priority of device. (default: 6)
SET IMP CTL=value	Sets the Unibus adapter number. (default: 3)
SHOW IMP ARP	Displays the IP address to MAC address table.

The IMP device must be attached to an available Ethernet interface. To determine which devices are available use the “SHOW ETHERNET” to list the potential interfaces. Check out the “0readme_ethernet.txt” from the top of the source directory.

The IMP device can be configured in several ways. Ether it can connect directly to an ethernet port (via TAP) or it can be connected via a TUN interface. If configured via TAP

interface the IMP will behave like any other ethernet interface and if asked grab it's own address. In environments where this is not desired the TUN interface can be used. When configured under a TUN interface the simulated network is a collection of ports on the local host. These can be mapped based on configuration options, see the 0readme_ethernet.txt file as to options.

With the IMP interface, the IP address of the simulated system is static, and under ITS is configured into the system at compile time. This address should be given to the IMP with the "set imp host=<ip>", the IMP will direct all traffic it sees to this address. If this address is not the same as the address of the system as seen by the network, then this address can be set with "set imp ip=<ip>", and "set imp gw=<ip>", or "set imp dhcp" which will allow the IMP to request an address from a local DHCP server. The IMP will translate the packets it receives/sends to look like the appeared from the desired address. The IMP will also correctly translate FTP requests in this configuration.

When running under a TUN interface, IMP is on a virtual 10.0.2.0" network. The default gateway is "10.0.2.1", with the default IMP at "10.0.2.15". For this mode DHCP can be used.

3.8.2 CH11 Chaosnet interface

Chaos net was another methods of network access for ITS. Chaosnet can be connected to another ITS system or a VAX/PDP11 running BSD Unix or VMS. Chaosnet runs over UDP protocol under UNIX. You must specify a node number, peer to talk to and your local UDP listening port. All UDP packets are sent to the peer for further processing.

SET CH NODE=*n*	Set the Node number for this system.
SET CH PEER=*ddd.ddd.ddd.ddd:dddd*	Set the Peer address and port number.
SET CH ADDR=value	Sets the Unibus address of device. (default: 764140)
SET CH VECT=value	Sets the interrupt vector of device. (default: 270)
SET CH BR=value	Sets the interrupt priority of device. (default: 6)
SET CH CTL=value	Sets the Unibus adapter number. (default: 3)

The CH device must be attached to a UDP port number. This is where it will receive UDP packets from it's peer.

3.8.3 KMC11 communications controller.

The KMC11-A is a general purpose microprocessor that is used in several DEC products. The KDP is an emulation of one of those products: COMM IOP-DUP. The COMM IOP-DUP

microcode controls and supervises 1 - 16 DUP-11 synchronous communications line interfaces, providing scatter/gather DMA, message framing, modem control, CRC validation, receiver resynchronization, and address recognition. The DUP-11 lines are assigned to the KMC11 by the (emulated) operating system, but SimH must be told how to connect them.

3.8.4 DUP11 network interface

Dup11 devices are used by DECNET to link machines together. They are a synchronous serial interface. This is implemented by UDP to send packets to the remote machine.

SET DUP LINES=*n*	Sets number of DUP 11 lines supported current max is 2.
SET DUPn SPEED=bits/sec	Set speed of link, default of 0 means no restrictions.
SET DUPn CORRUPTION=factor	Specify corruption factor.
SET DUPn W3	Enable Reset option
SET DUPn NOW3	Disable reset option
SET DUPn W5	Enable A dataset control option.
SET DUPn NOW5	Disable A dataset control option.
SET DUPn W6	Enable A & B dataset control option.
SET DUPn NOW6	Disable A & B dataset control option.
SET DUP ADDR=value	Sets the Unibus address of device. (default: 760300)
SET DUP VECT=value	Sets the interrupt vector of device. (default: 570)
SET DUP BR=value	Sets the interrupt priority of device. (default: 5)
SET DUP CTL=value	Sets the Unibus adapter number. (default: 3)

The DUP device must be attached to a UDP port number. This is where it will receive UDP packets from it’s peer.

Attach

The communication line performs input and output through a TCP session (or UDP session) connected to a user-specified port. The ATTACH command specifies the port to be used as well as the peer address:

```
sim> ATTACH DUP0 {interface:}port{,UDP},Connect=peerhost:port
```

where port is a decimal number between 1 and 65535 that is not being used for other TCP/IP activities.

Specifying symmetric attach configuration (with both a listen port and a peer address) will cause the side receiving an incoming connection to validate that the connection actually comes from the connecction destination system.

A symmetric attach configuration is required when using UDP packet transport.

The default connection uses TCP transport between the local system and the peer. Alternatively, UDP can be used by specifying UDP on the ATTACH command.

For more connection options, view the Help attach command for the DUP.

4. Symbolic Display and Input

The KL10 simulator implements symbolic display and input. These are controlled by the following switches to the EXAMINE and DEPOSIT commands:

-v	Looks up the address via translation, will return nothing if address is not valid.
-u	With the -v option, used user space instead of executive space.
-a	Display/Enter ASCII data.
-p	Display/Enter packed ASCII data.
-c	Display/Enter Sixbit Character data.
-m	Display/Enter Symbolic instructions
	Display/Enter Octal data.

Symbolic instructions can be of the formats:

- Opcode ac,operand
- Opcode operands
- I/O Opcode device,address

Operands can be one or more of the following in order:

- Optional @ for indirection.
- + or – to set sign.
- Octal number
- Optional (ac) for indexing.

Breakpoints can be set at real memory address. The PDP10 supports 3 types of breakpoints. Execution, Memory Access and Memory Modify. The default is execution. Adding -R to the breakpoint command will stop the simulator on access to that memory location, either via fetch, indirection or operand access. Adding -w will stop the simulator when the location is

modified.

The simulator can load RIM files, SAV files, EXE files, Waits Octal DMP files, MIT SBLK files.

When instruction history is enabled, the history trace shows internal status of various registers at the start of the instruction execution.

PC	Shows the PC of the instruction executed.
AC	The contents of the referenced AC.
EA	The final computed Effective address.
AR	Generally the operand that was computed.
RES	The AR register after the instruction was complete.
FLAGS	The values of the FLAGS register before execution of the instruction.
IR	The fetched instruction followed by the disassembled instruction.

The PDP10 simulator allows for Memory reference and memory modify breakpoints with the -r and -w options given to the break command.

5. OS specific configurations

5.1 ITS

To run ITS the CPU must be “set cpu ITS”, this will enable the ITS pager.